DETAILED DESCRIPTION OF THE INVENTION

Cost-Optimized Hardware Embodiments and Component Variants

In various embodiments, the penile anatomical geometry profiling device may be implemented using alternative component selections and simplified hardware configurations to optimize manufacturing cost, power consumption, or device scalability while preserving core synchronized distance and motion acquisition functionality.

In certain embodiments, a single time-of-flight (ToF) sensor may be utilized in combination with high-resolution longitudinal motion tracking to reconstruct geometry through dense positional sampling rather than multi-angle radial sampling. In other embodiments, lower-cost multi-zone ToF modules may replace multiple discrete sensors to reduce component count while maintaining surface contour resolution.

Optical motion tracking may be implemented using commercially available optical flow sensors, low-cost surface displacement sensors, or simplified emitter-receiver displacement pairs. In certain embodiments, a single motion sensor may be used with algorithmic correction routines to compensate for minor lateral drift, thereby reducing hardware complexity.

Anti-spoofing and physiological verification may be implemented using simplified pressure sensing components or capacitive proximity sensors in lieu of more complex photoplethysmography modules, depending on desired cost-performance balance. Encryption and secure processing may be achieved using microcontrollers incorporating integrated cryptographic accelerators rather than discrete secure elements in cost-sensitive embodiments.

The housing structure may be manufactured using injection-molded thermoplastics, overmolded elastomers, or modular snap-fit assemblies designed to minimize assembly steps and component alignment variability. In some embodiments, sensor alignment features may be integrated directly into molded structural geometry to reduce calibration complexity.

Firmware and reconstruction algorithms may be optimized for execution on low-power microcontrollers or mobile companion devices to reduce onboard processing requirements. Sampling density, reconstruction resolution, and comparative modeling features may be selectively scaled according to hardware capability tiers.

The cost-optimized embodiments described herein do not alter the inventive concept of acquiring synchronized radial distance measurements and longitudinal motion data to generate a user-specific penile anatomical geometry profile. Variations in component selection, sensor count, processing location, or material configuration remain within the scope of the present disclosure so long as geometry profile generation functionality is preserved.